the Expectations Hypothesis, the Fisher Hypothesis is rejected in the data. Nevertheless, the risk premiums, both relative to the Expectations Hypothesis and the Fisher Hypothesis, must be closely related across bonds of similar maturities; a ten-year bond must move, by our smart whack-a-mole investors, in a manner similar to a bond of maturity of nine years.

Term structure models specify the types of risk premiums and how they change. Thus, they capture how the two-year Treasury yield can deviate from an average of current and future T-bill rates (deviations from the Expectations Hypothesis), and how the long yield differs from the real rate and expected inflation (deviations from the Fisher Hypothesis). These models have three ingredients:

1. The underlying risk factors,
2. How the short rate moves, and
3. How the short rate and factor risk premiums affect long-term bonds.

The dynamics of the short rate in (2) can be interpreted as modeling how the Fed sets (or is implied to set) the interest rates at the shortest end of the yield curve in terms of risk factors (inflation and output according to the Taylor rule). The risk premiums specified in (3) compensate investors for maturity or duration risk, monetary policy risk, uncertainty, macro risk coming from inflation and growth shocks, and other risks. The key result is that the risk premiums differ across maturities and across time.[18] The workhorse models of today are the class of *affine term structure models*, which embody time-varying volatility, macro and latent factors, and what turns out to be very important to match the dynamics of yields in data—time-varying risk premiums.[19]

Term structure models derive both the dynamics of the entire cross section of bonds across maturities relative to each other, that is, the term structure of interest rates at a given point in time and specify how the term structure of interest rates evolves across time. In the context of equations (9.2) and (9.3), term structure models specify how risk premiums change across maturities. In a term structure model, the ten-year risk premium and, hence, yield is a function of the underlying factors and must move in such a way that it does not move "too much" (in a fashion consistent with no arbitrage) relative to the nine-year yield. The same must be true for the nine-year yield relative to the eight-year yield, and so on. Thus, the term structure model naturally picks up the notion of strong systematic

[18] The no arbitrage assumption in term structure models implies that investors receive the same *risk-adjusted* compensation across all bonds.

[19] The affine class of term structure models was formalized by Duffie and Kan (1996), but the first model was a pure Gaussian model developed by Vasicek (1977). Oldrich Vasicek played an important role in the development of the index fund industry (see chapter 17). A famous heteroskedastic model, also a special case of the affine framework, is Cox, Ingersoll, and Ross (1985). Time-varying risk premiums are essential in matching deviations from the Expectations Hypothesis, as Dai and Singleton (2002) show. For a summary of term structure models, see Piazzesi (2010).